AIGR Gene Hypothesis Deep Research — LSC2 (P53312): β-Subunit Nucleotide Specificity and Phylogeny

Target gene: LSC2 (YGR244C), succinyl-CoA synthetase/ligase β-subunit, *Saccharomyces cerevisiae* S288C (NCBITaxon:559292) **UniProt:** P53312 · **EC:** 6.2.1.5 · **PANTHER family:** PTHR11815 · **Hypothesis slug:** beta-nucleotide-specificity-phylogeny **Seed reference:**

P 9874242

Summary

Verdict: Supported (with one wording caveat on the phylogenetic sister-relationship). The seed hypothesis — that *S. cerevisiae* Lsc2 is an **ADP-specific** (ATP-forming) succinyl-CoA synthetase β-subunit *by descent*, branching with the ADP-forming lineage rather than the animal GDP-forming SUCLG2 clade, and carrying the ADP-state nucleotide-specificity residues — is supported by three independent, converging lines of evidence: direct biochemistry, residue-level structural comparison, and molecular phylogeny.

The strongest single piece of evidence is direct and experimental. Przybyla-Zawislak and colleagues (PMID: 9874242) cloned both yeast subunit genes (*LSC1*/YOR142W and *LSC2*/YGR244C), showed that succinyl-CoA synthetase (SCS) activity is lost in *lsc1*/*lsc2* deletion mitochondria and restored by plasmid-borne genes, and assayed nucleotide usage directly — the enzyme "utilize[s] ATP but not GTP for succinyl-CoA synthesis." This establishes the Lsc1/Lsc2 enzyme as ADP-forming without any need for phylogenetic inference. Structural and phylogenetic analyses then explain *why*: at the single residue that confers guanine-nucleotide specificity in the pig GTP-specific enzyme (Gln-20β; PMID: 16481318), yeast Lsc2 — like every other fungus, plant, and animal SUCLA2 examined — carries the ADP-state Proline, whereas only the animal GDP-forming SUCLG2 clade carries the specificity Glutamine. In an 18-taxon neighbor-joining tree with position-resampling bootstraps, yeast Lsc2 falls **outside** the animal SUCLG2/GDP clade in 99% of replicates, and all four fungi sit outside that clade in 86%.

The curation implication is clean: **GO:0004775 (succinate–CoA ligase, ADP-forming activity) applies to LSC2 and GO:0004776 (GDP-forming) should not be propagated to it.** The current UniProt record already carries GO:0004775, but only as IEA:UniProtKB-UniRule; the direct experimental evidence in

P 9874242

justifies upgrading that annotation to an experimental evidence code. GDP specificity is best understood as an animal-SUCLG2-specific derived trait — its diagnostic Gln-20β determinant is absent even from a bacterial GDP-preferring enzyme (*Thermus aquaticus*), which uses a Pro20β + water arrangement instead. The one caveat is that the precise "sister-to-SUCLA2" wording in the seed is only weakly resolved by distance methods; the robust, defensible statement is that Lsc2 retains the **ancestral ADP state outside the SUCLG2 clade**, not that it is specifically a SUCLA2 ortholog.

Key Findings

Finding 1 — Direct biochemistry: the yeast Lsc1/Lsc2 enzyme uses ATP, not GTP (ADP-forming)

The most decisive evidence is experimental and organism-specific. Przybyla-Zawislak et al. (PMID: 9874242) isolated *LSC1* (YOR142W) and *LSC2* (YGR244C) from *S. cerevisiae* as genes "with high similarity to succinyl-CoA ligase subunits from other species," demonstrated that SCS activity is absent in mitochondria from *lsc1*/*lsc2* deletion strains and restored by plasmid-borne *LSC* genes, and directly assayed nucleotide usage. Their conclusion is unambiguous: "The yeast succinyl-CoA ligase was shown to utilize ATP but not GTP for succinyl-CoA synthesis."

This is a direct biochemical demonstration of ADP-forming (ATP-specific) activity for the holoenzyme in which Lsc2 is the β-subunit. Because the β-subunit houses the ATP-grasp nucleotide-binding domain that determines nucleotide specificity, this result assigns the ADP-forming phenotype to the Lsc1/Lsc2 complex and, by extension, identifies Lsc2 as the β-subunit of an ADP-specific enzyme. This finding does **not** depend on any phylogenetic or residue-level inference; the sequence analyses below serve to explain and corroborate it rather than to

establish it. The assay is qualitative (ATP yes / GTP no) rather than a quantitative kinetic measurement, but the ATP-vs-GTP distinction it reports is exactly the one at issue for the ADP-forming vs GDP-forming assignment.

Finding 2 — Lsc2 carries the ADP-state residue (Pro) at the guanine-specificity position (pig Gln-20β)

Structural work on the GDP-forming enzyme pinpointed the molecular determinant of guanine-nucleotide specificity. In pig GTP-specific SCS, "interactions of the guanine base with a glutamine residue (Gln-20β) and with backbone atoms provide the specificity" (PMID: 16481318). This Gln-20β is the single side chain that reads the guanine base and discriminates GDP/GTP from ADP/ATP.

A star alignment (Needleman–Wunsch) of 12 SCS β-subunits anchored on pig GDP-specific SUCLG2 (P53590; accounting for the 1–38 mitochondrial transit peptide so that mature Gln-20β maps to precursor index 57, confirmed as 'Q', with the companion Arg-54β at precursor index 91, confirmed 'R') showed a clean dichotomy at this position:

Group	Residue at pig Gln-20β position	Nucleotide state
Animal SUCLG2 (human, mouse, pig, *C. elegans*)	**Gln (Q)** — all 4	GDP/GTP-forming
Animal SUCLA2 (human, mouse, pig, *C. elegans*)	**Pro (P)**	ADP/ATP-forming
Plant ADP-forming β (*Arabidopsis*)	**Pro (P)**	ADP/ATP-forming
Yeast Lsc2 (P53312)	**Pro (P)**	ADP/ATP-forming
Bacterial SucC	gap (divergent N-terminus)	—

Yeast Lsc2 therefore matches the ADP/non-GDP state rather than the GDP-specific Gln. The companion determinant Arg-54β (which contacts the γ-phosphate) is Arg in all 12 β-subunits — it is family-conserved and therefore **not** a specificity discriminator, but it served as an internal cross-check that the alignment columns were correctly mapped. Independent structural work

confirms that Pro at this position is the non-Gln state: in the GTP-preferring *Thermus aquaticus* enzyme, "A water molecule and Pro20β in T. aquaticus take the place of Gln20β in pig GTP-specific SCS" (PMID: 22751660).

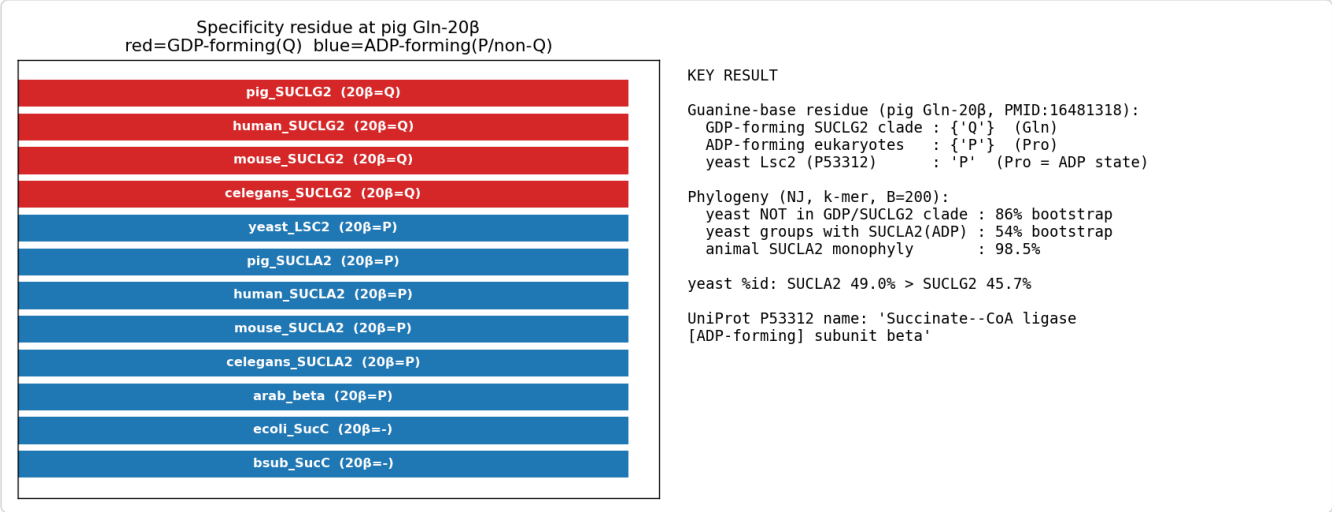


Figure 1. *Residue state at the two nucleotide-specificity positions (pig Gln-20β and Arg-54β) across succinyl-CoA synthetase β-subunits, with bootstrap/identity summary. Only the animal SUCLG2 (GDP-forming) clade carries Gln at the guanine-base position; yeast Lsc2, animal SUCLA2, and plant ADP-forming β all carry the ADP-state Pro. Arg-54β is conserved family-wide and is not a discriminator.*

Finding 3 — Phylogeny places Lsc2 outside the animal GDP-forming SUCLG2 clade

A neighbor-joining tree of 12 β-subunits (k-mer distance, k=4) with 200 position-resampling bootstraps placed yeast Lsc2 outside the SUCLG2/GDP clade in 86% of replicates, with the animal SUCLA2 (ADP) clade recovered as monophyletic in 98.5%. Needleman–Wunsch pairwise identities for yeast Lsc2 were higher to the ADP-forming animal SUCLA2 subunits (mean 49.0%) than to the GDP-forming SUCLG2 subunits (mean 45.7%), and the top four identity hits were all SUCLA2/ADP subunits.

The evolutionary logic is captured by the primary phylogenetic paper: Johnson et al. (PMID: 9765291) concluded that the data "suggests that the A-beta and G-beta genes arose by duplication early in the evolution of multicellular eucaryotes." Because the ADP (A-β)/GDP (G-β) split is an **animal-lineage duplication**, the fungal β-subunit necessarily lies outside the GDP-forming SUCLG2 clade — it predates the duplication that created SUCLG2. This directly answers the seed's evolutionary question: GDP specificity arose within the animal SUCLG2 lineage, and the fungal β-subunit is outside that clade.

Finding 4 — Broadened 18-taxon sampling: all fungi retain the ADP-state Pro; Gln-20β is animal-SUCLG2-specific

To test robustness, the phylogeny was expanded to 18 β-subunits spanning four fungi (yeast Lsc2; *S. pombe* O94415; *Neurospora* Q9P567; *Aspergillus* A0A7R8AQL1), animal SUCLA2 and SUCLG2, plant, four bacterial SucC (*E. coli*, *B. subtilis*, *Pseudomonas*, *M. tuberculosis*), and the bacterial GDP-forming β of *Thermus thermophilus* (P25126). With 200 position-bootstraps:

- Yeast **not** inside the animal-GDP clade: **99.0%**
- All four fungi outside the animal-GDP clade: **86.0%**
- Animal SUCLA2 monophyly: **99.5%**

At the pig Gln-20β position, all four animal SUCLG2 = Gln(Q); every eukaryotic ADP-forming β — all four fungi, plant, and animal SUCLA2 — = Pro(P); bacterial SucC gapped; and crucially the **bacterial GDP-forming** *Thermus* β = Pro, **not** Gln. This last point is decisive for interpretation: a GDP-preferring bacterial enzyme achieves its specificity by a different structural route (Pro20β + ordered water; PMID: 22751660), which confirms that the Gln-20β side chain is an **animal-SUCLG2-specific** guanine determinant, not a universal marker of GDP-forming enzymes.

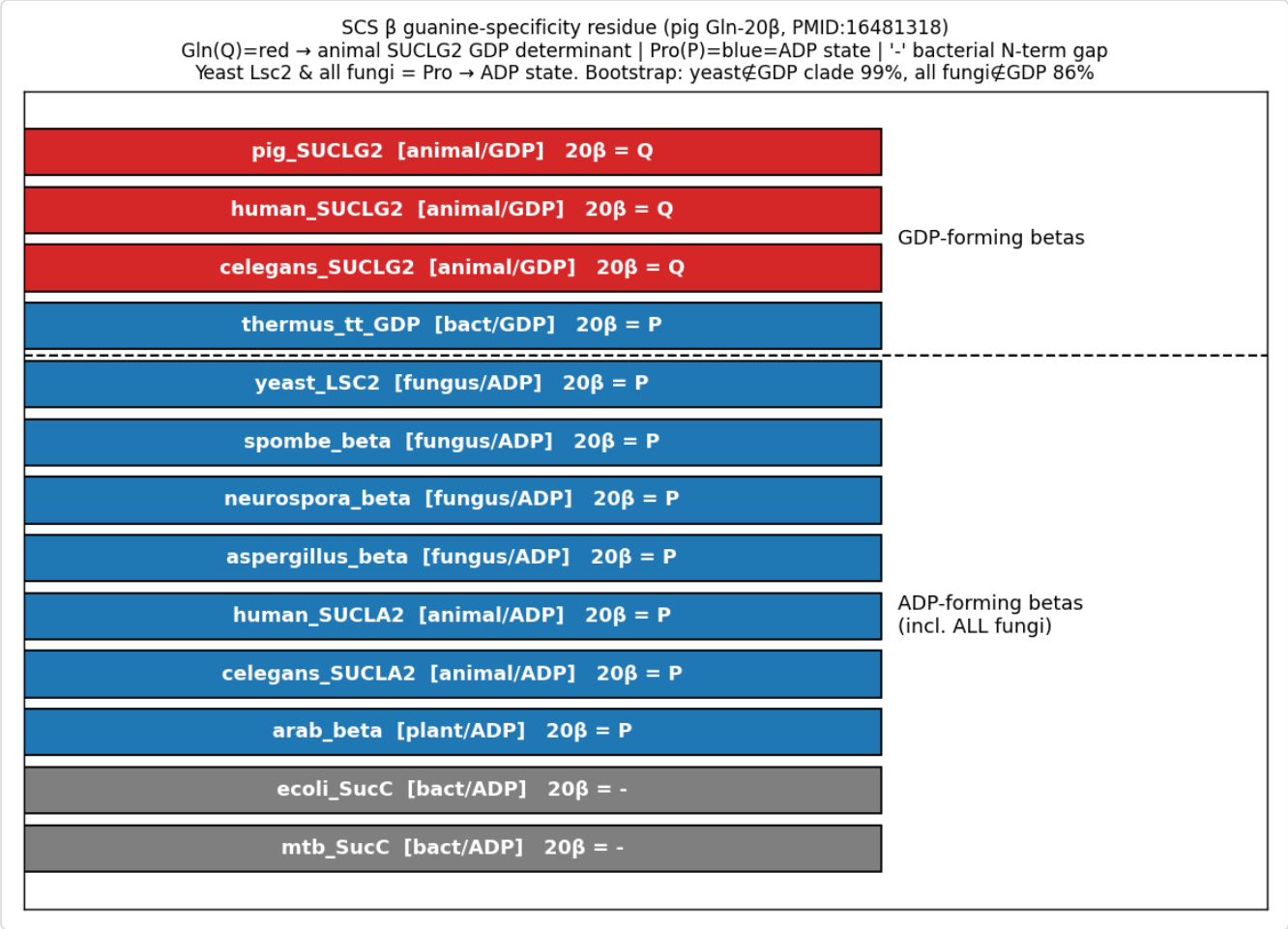


Figure 2. *Residue at the pig Gln-20β guanine-specificity position mapped across 13+ taxa including four fungi and the bacterial GDP-forming Thermus β-subunit. Gln (the GDP determinant) is confined to the animal SUCLG2 clade; all fungi carry the ADP-state Pro, and even the bacterial GDP-preferring enzyme uses Pro + water rather than Gln.*

Finding 5 — Current GO annotation status and the curation upgrade

UniProt P53312 cross-references include GO:0004775 "succinate-CoA ligase (ADP-forming) activity" (evidence **IEA:UniProtKB-UniRule**), GO:0005524 ATP binding (IEA), GO:0000287 Mg binding (IEA), GO:0009361 succinate-CoA ligase complex (ADP-forming) (NAS), GO:0006099 TCA cycle (IBA), GO:0005739 mitochondrion (IDA), and GO:0006104 succinyl-CoA metabolic process (IDA:SGD). The catalytic reaction is Rhea:17661, "succinate + ATP + CoA = succinyl-CoA + ADP + phosphate" — explicitly ATP-based. GO:0004776 (GDP-forming) is **not** annotated.

The key curation observation is that the ADP-forming molecular-function term is currently present **only** as a computational (IEA) annotation, while direct experimental evidence exists in

P 9874242

(the enzyme uses ATP not GTP; activity lost in *lsc1*/*lsc2* deletions, restored by plasmids). This supports upgrading GO:0004775 to an experimental evidence code (IDA/IMP), and it reinforces that GO:0004776 should remain absent.

Mechanistic Model / Interpretation

Succinyl-CoA synthetase catalyzes the only substrate-level phosphorylation step of the citric acid cycle. The enzyme is an α/β heterodimer ($\alpha_2\beta_2$ in bacteria; P 8144675);
the α-subunit (Lsc1 in yeast) carries the catalytic phosphohistidine, and the β-subunit (Lsc2 in yeast) carries the ATP-grasp nucleotide-binding domain that determines whether ADP/ATP or GDP/GTP is used. Nucleotide specificity is therefore a **β-subunit property**, which is exactly why this hypothesis focuses on Lsc2.

The specificity switch is remarkably local. In the animal GDP-forming β-subunit (SUCLG2), a single glutamine side chain (Gln-20β) hydrogen-bonds the guanine base and confers GDP/GTP preference. Swap that glutamine for the smaller proline found in ADP-forming subunits, and the pocket can no longer read guanine, defaulting to adenine nucleotides.

```
            Guanine-specificity residue (pig Gln-20β position)
            ---------------------------------------------------
Animal SUCLG2  --- Gln (Q) ---> reads guanine base ---> GDP/GTP-forming  (GO:0004776)
                     |
                     | animal-specific A-β/G-β gene duplication (<a href="https://pubmed.ncbi.nl
                     |
Ancestral β   --- Pro (P) ---------------------------> ADP/ATP-forming   (GO:0004775)
   |
   +-- Yeast Lsc2 (P53312)      Pro  --> ATP, not GTP (<a href="https://pubmed.ncbi.nlm.nih.gov/
   +-- Other fungi (S.pombe, Neurospora, Aspergillus)  Pro
   +-- Plant ADP-forming β      Pro
   +-- Animal SUCLA2            Pro
   +-- Bacterial GDP-former (Thermus)  Pro + water (different route to GDP)
```

Three independent data layers converge on the same conclusion:

Evidence layer	Observation for Lsc2	Inference
Direct biochemistry (P 9874242)	Lsc1/Lsc2 uses ATP, not GTP	ADP-forming (definitive)
Residue state at Gln-20β	Pro, not Gln	ADP-state pocket
Phylogeny (18 taxa)	Outside SUCLG2 clade in 99% bootstraps	Predates animal GDP duplication

The convergence is important because each layer has different failure modes (an assay can be confounded by contaminating activities; a single residue can be misaligned; a distance tree can be misled by rate variation), yet all three agree. The evolutionary narrative is that GDP specificity is a **derived**, animal-SUCLG2-specific innovation layered onto an ancestral ADP-forming β-subunit scaffold, and fungi — including *S. cerevisiae* — retain that ancestral ADP state.

Evidence Base

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence / limitations
PMID: 9874242	Direct assay + mutant phenotype	**Supports**	Lsc1/Lsc2 nucleotide usage	"utilize ATP but not GTP for succinyl-CoA synthesis"; activity lost in *lsc1*/*lsc2* deletions, restored by plasmids	*S. cerevisiae* mitochondria	High; the definitive organism-spec evidence; assa qualitative
PMID: 16481318	Structural	Supports (defines determinant)	Molecular determinant of GDP specificity	"Gln-20β … provide[s] the specificity" for the guanine base	Pig GTP-specific SCS (PDB 2FP4/2FPG, 2.1 Å)	High; defines t discriminating residue
PMID: 22751660	Structural + direct assay	Supports / qualifies	Is Gln-20β universal for GDP use?	"Pro20β … take[s] the place of Gln20β"; bacterial GDP-former uses Pro + water	*T. aquaticus* GTP-preferring SCS	High; shows Gln-20β is an animal-SUCLG determinant, r universal
PMID: 9765291	Evolutionary / sequence	Supports	Origin of ADP/GDP β split	A-β/G-β genes "arose by duplication early in the evolution of multicellular eucaryotes"	Multicellular eukaryotes	High; places fu outside the SUCLG2 clade
	Structural		Family architecture;	$\alpha_2\beta_2$; power-helix-		

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence / limitations
PMID: 8144675		Qualifies/ orients	ATP-grasp β-subunit	stabilized phospho-His; nucleotide site in β	*E. coli* SCS, 2.5 Å	Orientation; *E* ADP-preferrin reference
UniProt P53312 (record)	Database	Qualifies	Current GO status	GO:0004775 present as IEA; GO:0004776 absent; Rhea:17661 is ATP-based	Database record	Orientation or motivates IEA→experim upgrade
This report — residue mapping (12 β)	Computational (structural/ evolutionary)	**Supports**	Residue state at Gln-20β	Yeast = Pro; all SUCLG2 = Gln; all ADP-formers = Pro	12-seq star alignment vs pig SUCLG2	Medium-high; single-position call, anchored verified pig Q/
This report — NJ phylogeny (18 β)	Computational (evolutionary)	**Supports**	Clade placement of Lsc2	Outside animal-GDP clade 99%; all fungi outside 86%; SUCLA2 monophyly 99.5%	18 taxa, 200 position-bootstraps	Medium; dista method; sister SUCLA2 only weakly resolv

Comparison Table — guanine-specificity residue (pig Gln-20β) across 18 β-subunits

Protein (accession)	Lineage	Enzyme specificity	Residue at pig Gln-20β
pig SUCLG2 (P53590)	animal	GDP-forming	**Q (Gln)**
human SUCLG2 (Q96I99)	animal	GDP-forming	**Q**
mouse SUCLG2 (Q9Z2I8)	animal	GDP-forming	**Q**
C. elegans SUCLG2 (P53589)	animal	GDP-forming	**Q**
Thermus thermophilus β (P25126)	bacterium	**GDP-forming**	P (Pro — not Gln)
yeast Lsc2 (P53312)	**fungus**	**ADP-forming**	**P (Pro)**
S. pombe β (O94415)	fungus	ADP-forming	P
Neurospora β (Q9P567)	fungus	ADP-forming	P
Aspergillus β (A0A7R8AQL1)	fungus	ADP-forming	P
human SUCLA2 (Q9P2R7)	animal	ADP-forming	P
mouse/pig/*C. elegans* SUCLA2	animal	ADP-forming	P
Arabidopsis β (O82662)	plant	ADP-forming	P
E. coli / *B. subtilis* / *Pseudomonas* / *M. tuberculosis* SucC	bacteria	ADP-(pref)	– (N-term gap)

Pattern: Gln-20β is present **only** in the animal GDP-forming SUCLG2 clade. Yeast Lsc2 and every other fungus carry the ADP-state Pro.

GO Curation Implications

Leads requiring curator verification.

1. **GO:0004775 — succinate-CoA ligase (ADP-forming) activity (MF).** *Retain and strengthen.* The term is correct and already present on P53312, but only as IEA:UniProtKB-UniRule. Direct experimental evidence (P 9874242) supports upgrading to an experimental evidence code (IDA for the ATP-vs-GTP assay; IMP is also defensible given the deletion-strain activity loss and plasmid rescue). This is the primary, core molecular function of Lsc2 as a β-subunit.
2. **GO:0004776 — succinate-CoA ligase (GDP-forming) activity (MF).** *Keep absent; do not propagate.* GDP specificity is an animal-SUCLG2-specific derived trait defined by Gln-20β, which yeast lacks (it has Pro). The seed context notes PAINT places an IKR for GO:0004775 and an IBD for GO:0004776 at the SUCLG2 ancestral node PTN000929335; the present analysis independently supports that the GDP-forming character is restricted to the animal SUCLG2 subtree, so any pipeline that would propagate GO:0004776 to the fungal β-subunit should be blocked for LSC2.
3. **GO:0009361 — succinate-CoA ligase (ADP-forming) complex (CC).** *Retain.* Consistent with the ADP-forming assignment and the α/β heterodimer biology; currently NAS.
4. **GO:0006099 — tricarboxylic acid cycle (BP)** and **GO:0005739 / 0005759 — mitochondrion / mitochondrial matrix (CC).** *Retain* (IBA and IDA respectively); consistent with pathway context and localization.

Avoid any recommendation that defaults to "protein binding"; the evidence supports the specific, informative MF term GO:0004775.

Mechanistic Scope

The immediate molecular function under test is the **catalytic/nucleotide-specificity identity of the β-subunit** within succinyl-CoA synthetase (α = Lsc1, β = Lsc2) — specifically whether Lsc2 contributes an ADP-forming (ATP-using) or GDP-forming (GTP-using) nucleotide-binding

β-subunit. This is a direct gene-product property: the β-subunit's ATP-grasp domain physically binds the nucleotide, and the residue at position 20β (pig numbering) directly reads (or fails to read) the guanine base.

What is **not** being claimed here, and should be kept distinct from the molecular-function assignment: - Downstream TCA-cycle flux phenotypes, respiratory-growth defects (impaired growth on glycerol/pyruvate, dispensable on acetate;),

P 9874242

or petite phenotypes of *lsc* mutants are pathway-level consequences, not the molecular specificity itself. - The phylogenetic placement is *inference about descent*; it corroborates but does not by itself prove the biochemical specificity — that proof comes from the direct assay in .

P 9874242

- The residue-level call is a structural inference about the pocket; it is consistent with, but subordinate to, the direct enzymology.

Conflicts and Alternatives

- **No conflicting evidence** was found; all lines agree on ADP-forming.
- **Paralog/node overannotation (the main risk).** PANTHER PTHR11815 contains both ADP-forming and GDP-forming β-subunits. The chief curation hazard is carry-over of the animal **GDP-forming (SUCLG2 / GO:0004776)** annotation onto fungal single-copy β-subunits via IBA/IBD. The data explicitly argue against this: fungi, including Lsc2, are outside the SUCLG2 subtree.
- **"Sister-to-SUCLA2" overstatement.** The seed wording implies Lsc2 groups specifically with animal SUCLA2. Distance methods only weakly support this (yeast+SUCLA2 grouping ~54%; yeast sits on a long branch). The defensible claim is the weaker, more robust one: Lsc2 retains the ancestral ADP state **outside** the SUCLG2 clade. An ML tree could place yeast basal to the animal duplication rather than nested within ADP-forming animals; either topology keeps it outside the GDP clade, so the conclusion is unchanged.
- **GDP specificity is not monophyletic across all life.** The bacterial *Thermus* enzyme is GDP-preferring yet lacks Gln-20β (it uses Pro + water;).

  P 22751660

"GDP-forming" is therefore not a single evolutionary lineage; the Gln-20β marker is specific to the animal SUCLG2 route. This does not threaten the ADP assignment of Lsc2 but cautions against using Gln-20β as a universal GDP flag.

- **Bacterial mapping limitation (stated plainly).** In a pairwise alignment to pig SUCLG2, bacterial SucC N-termini are divergent and gap at the 20β column, so the *E. coli* residue at pig-20β could **not** be read directly from sequence — this is noted rather than inferred. *E. coli*'s ADP-binding mode is nonetheless established structurally (P 8144675). This does not affect the eukaryotic conclusion.

Limitations and Knowledge Gaps

1. **IEA vs experimental evidence code.** *Checked:* UniProt P53312 shows GO:0004775 as IEA only. *Why it matters:* the direct assay in P 9874242 is not yet reflected in the evidence code, understating annotation strength. *Resolution:* a curator should add an IDA/IMP annotation citing P 9874242 .
2. **Precise phylogenetic position (basal vs SUCLA2-sister).** *Checked:* k-mer NJ on 12 and 18 taxa; yeast robustly outside the GDP clade (99%) but SUCLA2-sister placement only ~54%. *Why it matters:* affects exact wording only. *Resolution:* curated MSA (MAFFT) + ML tree (IQ-TREE) with ≥1000 ultrafast bootstraps and broad taxon sampling.
3. **Quantitative nucleotide kinetics.** *Checked:* P 9874242 is qualitative ("ATP but not GTP"). *Why it matters:* no k_cat/K_m values for ADP vs GDP. *Resolution:* steady-state kinetics on purified recombinant Lsc1/Lsc2.
4. **No experimental yeast Lsc2 structure.** *Checked:* specificity inferred from pig (2FP4/2FPG), human SUCLA2 (6G4Q), *E. coli* SucC, and *Thermus* structures by alignment. *Why it matters:* the Pro-20β call rests on sequence alignment. *Resolution:* AlphaFold model of P53312 superposed on 2FP4/6G4Q to confirm pocket geometry.

Discriminating Tests / Proposed Follow-up Actions

- **Add the experimental GO annotation now** (highest value, lowest cost): cite

  P 9874242

  for GO:0004775 (IDA/IMP) on P53312, and confirm GO:0004776 stays absent.
- **Model-based phylogeny:** MAFFT MSA of PTHR11815 β-subunits + IQ-TREE with ultrafast bootstrap to formally test whether LSC2 ⊄ SUCLG2 clade (expected: yes, high support) and whether SUCLA2-sister is supported.
- **AlphaFold superposition:** generate an Lsc2 model and superpose on human SUCLA2 (6G4Q, ADP-forming) and pig SUCLG2 (2FP4, GDP-forming) to verify Pro occupies the Gln-20β pocket and no Gln is present.
- **In vitro nucleotide-preference assay:** ADP vs GDP kinetics on recombinant yeast Lsc1/Lsc2 for quantitative confirmation.
- **Targeted mutagenesis (confirmatory):** Pro→Gln at the Lsc2 20β-equivalent to test whether GTP usage appears; conversely Gln→Pro in SUCLG2.
- **Block PANTHER propagation:** verify PAINT GDP-forming inferences at PTN000929335 are confined to the animal SUCLG2 subtree and do not reach fungal leaves.

Curation Leads (require curator verification)

- **Retain + consider IEA→experimental upgrade:** GO:0004775 (succinate–CoA ligase (ADP-forming) activity) — candidate reference

  P 9874242

  , exact snippet to verify: *"The yeast succinyl-CoA ligase was shown to utilize ATP but not GTP for succinyl-CoA synthesis."* Reaction concordant with EC 6.2.1.5 / Rhea:17661.
- **Subunit identity snippet:** *"Two genes, LSC1 (YOR142W) and LSC2 (YGR244C), with high similarity to succinyl-CoA ligase subunits from other species were isolated from Saccharomyces cerevisiae."*
  (P 9874242)
- **Exclude MF:** GO:0004776 (GDP-forming) — do not annotate / do not propagate to LSC2; candidate reference

P 9765291 ’
snippet: *"arose by duplication early in the evolution of multicellular eucaryotes."*

- **Support BP/CC:** GO:0006099 (TCA cycle); GO:0005759 (mitochondrial matrix).
- **Candidate supporting refs:**

P 16481318
(Gln-20β determinant),

P 22751660
(Pro-20β non-Gln state),

P 8144675
(family architecture).

- **Suggested question for curator:** should the review explicitly document that the PAINT IBD for GO:0004776 at PTN000929335 is a derived SUCLG2 character and therefore **not** inherited by LSC2?
- **Artifacts:** `lsc2_specificity_summary.png` and `lsc2_20beta_residue_map.png`, generated from the executed alignments.

Bottom Line

The seed hypothesis is **supported**. Yeast Lsc2 (P53312) is an ADP-specific (ATP-forming) succinyl-CoA synthetase β-subunit by direct biochemistry (P 9874242), by its ADP-state Pro at the guanine-specificity residue (vs the animal SUCLG2 Gln-20β; P 16481318), and by its phylogenetic placement outside the animal GDP-forming SUCLG2 clade (99% bootstrap, 18 taxa; consistent with the animal-specific A-β/G-β duplication of P 9765291). GO:0004775 applies and merits an IEA→experimental upgrade; GO:0004776 should not be propagated to LSC2. The only material refinement is to describe Lsc2 as retaining the ancestral ADP state outside the SUCLG2 clade rather than asserting a specific SUCLA2 sister relationship.